DEVIATION REPORT

Out-of-specification result - Atorvastatin Calcium USP, batch AT-24118

1. IDENTIFICATION

Deviation No: DEV-2026-0142

Date of occurrence: 09-Oct-2026

Date detected: 12-Oct-2026

Detected by: Anita Rao, QC Analyst

Department: Quality Control

Site / Area: Unit 2, Building B, Drying Room DR-04

Product: Atorvastatin Calcium USP (API), material code API-ATV-02

Batch No: AT-24118 Batch size: 250 kg

Process stage: Vacuum tray drying (final unit operation before milling)

Equipment: Vacuum tray dryer VD-02; RTD temperature probe TT-207

Classification (proposed): Major

Deviation type: Unplanned

2. DESCRIPTION

During QC release testing on 12-Oct-2026, batch AT-24118 failed the specification for the specified impurity desfluoro atorvastatin at 0.42% w/w against a limit of not more than 0.30% w/w. Assay was 97.1% w/w against 98.0-102.0% w/w. Total impurities were 0.71% w/w against a limit of not more than 1.0% w/w; all unspecified impurities were within limits.

The batch was dried in VD-02 from 02:10 to 11:40 on 09-Oct-2026 (9 h 30 min). The BPR limit is not more than 8 h at 65-70 C. Controller temperatures recorded every 2 h were 66.2, 67.1, 66.8, 67.4 and 67.0 C.

At 06:40 on 09-Oct-2026 the night-shift operator recorded a handheld reference thermometer reading of 79.4 C at the jacket outlet and noted: "higher than controller, controller reading OK, continued".

3. IMMEDIATE ACTION / CONTAINMENT

- Batch AT-24118 and retain samples quarantined under QA hold QA-H-2026-118.
- VD-02 blocked for use and campaign drying stopped.
- QC initiated an OOS investigation under SOP QC-OOS-002.
- No product from this batch has been released or distributed.

4. IMPACT ASSESSMENT

No patient exposure: the batch has not been released. Desfluoro atorvastatin is a known thermal degradant of atorvastatin. Batches dried on VD-02 after the last calibration date may be affected. AT-24116 and AT-24117, manufactured from the same API lot and the same recovered solvent lot, passed release.

5. RECORDS REVIEWED (ATTACHMENTS)

1. BPR AT-24118, PAGES 1-14, INCLUDING DRYING LOG PAGE 7 AND HANDHELD VERIFICATION PAGE 9.

2. QC RELEASE DATA PACKAGE AT-24118, INCLUDING HPLC SYSTEM SUITABILITY: RESOLUTION 2.4, TAILING 1.08.

3. CALIBRATION CERTIFICATE CAL-2025-8841 FOR RTD TT-207: LAST CALIBRATED 15-SEP-2025, NEXT DUE 15-SEP-2026.

4. CMMS WORK-ORDER EXPORT FOR TT-207: NO CALIBRATION WORK ORDER RAISED AFTER 12-MAR-2026.

5. CHANGE CONTROL CC-2026-041, CMMS V6 TO V8 MIGRATION, LIVE 01-AUG-2026.

6. SOP QA-SOP-114 REV 4, DRYING OPERATION.

7. TRAINING RECORD FOR OPERATOR R. PATIL, TRAINED 18-NOV-2024, NO REFRESHER SINCE.

8. DEVIATION HISTORY FOR VD-02: DEV-2025-031 AND DEV-2026-009.

9. CALIBRATION PROGRAMME REVIEW FOR CC-2026-041: 14 INSTRUMENTS MIGRATED; 3 WITHOUT WORK ORDERS (TT-207, PT-114, FT-208).

10. COA REVIEW: PRECURSOR LOT RM-26-0912 AND RECOVERED TOLUENE LOT SOL-26-077 BOTH WITHIN SPECIFICATION.

6. INITIAL OBSERVATIONS

- TT-207 was 24 days past its calibration due date when the batch was executed.
- The BPR records the handheld check but has no field requiring a decision when the reading differs from the controller.
- Both previous VD-02 temperature deviations were closed with operator counselling; no calibration or SOP review was performed.
- The CC-2026-041 impact assessment explicitly excludes maintenance plan and calibration schedule validation.

7. PROPOSED INVESTIGATION SCOPE

Drying step of AT-24118; calibration status of TT-207; all batches of any product dried on VD-02 since 15-Sep-2026; impact of the CMMS migration on calibration schedules for the 14 migrated instruments.

Prepared by: Anita Rao, QC Analyst 12-Oct-2026

Reviewed by: S. Kulkarni, QC Head 12-Oct-2026

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